

GRIDBENCH · REQUIREMENTS

GridBench requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/gridbench		
Source	docs/03-requirements.md @ daed09d		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (12 measurable requirements, status against the concept)
0.1	2026-09-25	Amish Chadha	Initial scaffold

GridBench requirements

Twelve requirements define a garage-built grid bench and fixture set. At concept stage, three are not met or are at risk: cost (R10), tipping stability without ballast (R12) and the long-span accuracy of the plywood field in humid seasons (R3). Status values are estimates from the design precis (GBN-PRC-001) and must be checked by calculation at TRL 3 and by test later.

■ **Requirements**

Table 1. GridBench requirements and concept status.

ID	Requirement	Target	Verification	Concept status
R1	Grid standard	25.00 mm pitch in X and Y, M6 threads, hole centers 12.5 mm from the worktop edges; fits metric optical breadboard accessories	Inspection of drawing; fit test with a commercial M6 accessory	Met by design
R2	Worktop size and height	At least 1,200 x 600 mm; top at 900 mm, adjustable ± 15 mm with levelling feet	Measurement	Met by design
R3	Coarse field hole position	± 0.3 mm hole to hole over 1,000 mm at 40 to 60 % RH (template-drilled build: ± 0.5 mm)	Measurement with steel rule and pin gauges	At risk: plywood moisture movement could reach about 0.4 mm over 1,200 mm (estimate)
R4	Precision tile location	Hole position ± 0.05 mm; a part relocated on two 8 mm dowels returns within 0.03 mm	Dial indicator test, 10 relocations	Unverified; depends on machining access (drill press with jig, or CNC)
R5	Flatness	Tile 0.05 mm over 300 mm; worktop 0.5 mm over 1,000 mm	Straightedge and feeler gauges	Tile met if bought as cast tooling plate; worktop unverified
R6	Stiffness and load	150 kg distributed load; deflection 0.5 mm or less under 500 N at the middle of a bay	Load test	Met by estimate (about 0.3 to 0.4 mm)
R7	Clamp hold-down	Each printed toe clamp holds 500 N at the part with no visible creep after 8 h at 25 °C	Spring scale pull test	Unverified (PETG creep)
R8	Fixture change	Swap one fixture for another in 60 s or less with one 5 mm hex key	Timed trial	Met by design
R9	Buildability	Built with hand and garage tools plus an FDM printer (200 x 200 mm bed); CNC optional	Build review	Met by design
R10	Parts cost	\$220 or less for bench, grid, tile, fixture set and indicator	Priced BOM	Not met: about \$239 (indicative)
R11	Openness	All geometry as build123d source; fixture library parametric; CERN-OHL-S-2.0	Repository review	Met by design
R12	Stability and edges	No tipping under 150 N horizontal at the top edge; all exposed edges chamfered or rounded 0.5 mm or more	Push test; inspection	Not met without ballast: about 105 N empty, about 160 N with 20 kg on the shelf (estimates)

■ Design load case

- Distributed load: 150 kg over the worktop (a fixture set, tools and parts).
- Point load: 500 N from a person leaning on a clamped part, at the middle of a bay between cross rails.

- Horizontal push: 150 N at the front edge of the worktop, as when pushing a tight part onto dowels or filing.

■ Assumptions

- Birch plywood with a modulus of about 7 GPa along the face grain (estimate, to be checked at TRL 3).
- Pine or equivalent softwood frame at about 500 kg/m³.
- Workshop humidity swings of about 5 % in wood moisture content between seasons.

■ Requirements not met

- **R10 cost:** about \$239 against \$220. Options are in the precis and the review note.
- **R12 stability:** the empty bench is too light to resist 150 N at the top edge. Ballast on the shelf or a wall anchor is needed.
- **R3 at risk:** the plywood field may drift with humidity over its full length. The precision tile carries all tight-tolerance work.